

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

18/20

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I K. Yamini (19257218) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

K. Y . .

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:

DataLink → MACAddress → LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→ DataLinkLayer → ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

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SLOT:B

COURSE NAME : COMPUTER NETWORKS

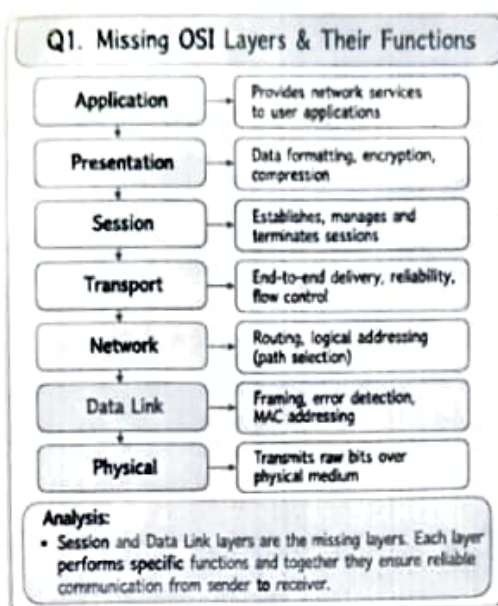
Comprehensive Analysis and Concept Mapping in Computer Networks

Q1. Missing OSI Layers & Their Functions

The **Open Systems Interconnection (OSI) Model** is a seven-layer reference framework developed to standardize communication between different computer systems. It divides the communication process into seven distinct layers, where each layer performs a specific function and provides services to the layer above it. In the given concept map, the missing layers are **Session Layer** and **Data Link Layer**.

The communication process begins at the **Application Layer**, where users access network services. The **Presentation Layer** translates, encrypts, and compresses data into a common format. The **Session Layer** establishes, manages, synchronizes, and terminates communication sessions between devices. It ensures continuous dialogue and enables recovery if communication is interrupted. The **Transport Layer** provides reliable end-to-end communication through segmentation, flow control, and error recovery. The **Network Layer** determines the best path for data transmission using logical addressing and routing. The **Data Link Layer** packages data into frames, performs error detection, controls media access, and ensures reliable communication between directly connected devices using MAC addresses. Finally, the **Physical Layer** transmits raw bits over the communication medium through electrical, optical, or radio signals.

Together, the **Session Layer** and **Data Link Layer** play a crucial role in ensuring reliable communication. The Session Layer maintains organized communication between applications, while the Data Link Layer detects transmission errors and delivers data accurately across the local network.



Concept Map Architecture (Q1)

Source Concept (Node A)	Linking Relationship	Target Concept (Node B)
Application Layer	provides user services through	Presentation Layer
Presentation Layer	translates and formats data before	Session Layer
Session Layer	establishes and manages communication for	Transport Layer
Transport Layer	provides reliable end-to-end delivery to	Network Layer
Network Layer	routes packets toward	Data Link Layer
Data Link Layer	frames data and performs error detection before	Physical Layer
Physical Layer	transmits raw bits over	Communication Medium

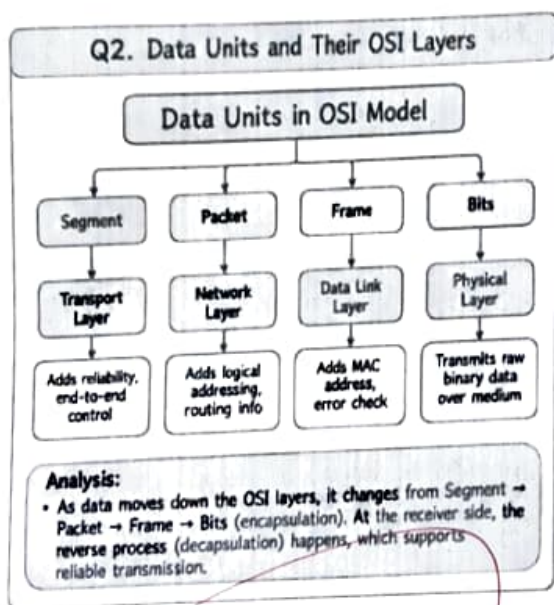
Literature Insights & Evidence (Q1)

- Andrew S. Tanenbaum & David J. Wetherall (2021): Explain the seven-layer OSI architecture and the role of each communication layer.
- Behrouz A. Forouzan (2022): Describes Session Layer services, Data Link Layer framing, error detection, and flow control.

- William Stallings (2020): Discusses reliable data communication, networking protocols, and the functions of OSI layers.
- Kurose & Ross (2023): Explain layered network architecture and reliable communication mechanisms.

Q2. Data Units and their OSI Layers

The **OSI (Open Systems Interconnection) model** organizes data communication into seven layers. As data travels from the sender to the receiver, it is transformed into different **Protocol Data Units (PDUs)** — **Data, Segment, Packet, Frame, and Bits**. Each transformation adds or removes protocol information (headers/trailers), ensuring reliable, accurate, and efficient transmission across networks.



Input	Process (OSI Layer)	Output (PDU)	Purpose
User/Application Data	Application, Presentation & Session Layers	Data	Creates, formats, and manages user data
Data	Transport Layer	Segment	Adds TCP/UDP header for reliable communication
Segment	Network Layer	Packet	Adds IP address for routing across networks
Packet	Data Link Layer	Frame	Adds MAC address and trailer for local delivery and error detection

Frame	Physical Layer	Bits	Converts the frame into binary signals (0s and 1s) for transmission.
bits	Transmission Medium	Signals	Transfers data through cable, fiber, or wireless media.
Signals	Receiver (Decapsulation)	Original Data	Removes headers layer by layer to recover the original user data.

Literature Insights & Evidence (Q2)

- The OSI model is a seven-layer framework that standardizes communication between different network devices by assigning specific functions to each layer.
- As data moves through the OSI layers, it undergoes encapsulation, transforming from Data → Segment → Packet → Frame → Bits, with each layer adding the required control information.
- At the receiver's end, decapsulation removes the headers and trailers in reverse order, restoring the original data for the application.
- Studies by Forouzan (2017), Kurose & Ross (2021), and Stallings (2020) show that the layered architecture improves reliability, interoperability, scalability, and efficient network communication.

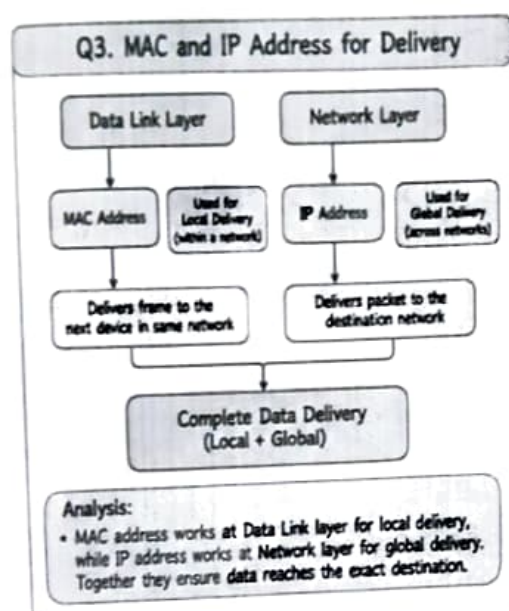
Q3. MAC and IP Address for Delivery

The communication process in computer networks relies on two complementary layers to ensure complete and reliable data delivery from a source device to a destination device.

At the **Data Link Layer**, communication is based on the **MAC (Media Access Control) Address**, which is a unique hardware address assigned to every network interface. The MAC address is responsible for **Local Delivery**, meaning it transfers data between devices that are connected within the **same local network (LAN)**. Switches use MAC addresses to forward frames accurately to the intended device, ensuring efficient communication inside the local network.

At the **Network Layer**, communication is based on the **IP (Internet Protocol) Address**, which uniquely identifies devices across different networks. The IP address enables **Global Delivery** by allowing routers to determine the best path and forward packets from one network to another until they reach the destination network. This makes communication possible over the Internet and other interconnected networks.

Both layers work together to achieve **end-to-end communication**. The **IP address** ensures that data travels across multiple networks to the correct destination network (global delivery), while the **MAC address** ensures that the data is delivered to the correct device within that local network (local delivery). Thus, **IP** provides routing between networks, and **MAC** provides final delivery within the network, ensuring complete, accurate, and reliable data transmission.



Concept	Address Used	Function	Description
Data Link Layer	MAC Address	Local Delivery	Uses the MAC address to deliver data frames between devices within the same Local Area Network (LAN).
Network Layer	IP Address	Global Delivery	Uses the IP address to route packets across different networks until they reach the destination network.
Complete Data Delivery	MAC Address + IP Address	End-to-End Communication	IP address enables global routing between networks, while MAC address ensures final delivery to the correct device in the local network.

Literature Insights & Evidence (Q3)

- Forouzan (2017) states that the Data Link Layer uses MAC addresses to provide reliable communication between devices within the same local network (LAN), ensuring accurate frame delivery.

- Tanenbaum & Wetherall (2011) explain that the Network Layer uses IP addresses to route packets across interconnected networks, enabling global communication through routers.
- Kurose & Ross (2021) emphasize that MAC addresses handle local delivery, while IP addresses enable end-to-end communication across different networks, with both working together for successful data transmission.
- The IEEE 802 standards define MAC addressing for local network communication, while the Internet Protocol (IP) standards established by the IETF provide the foundation for global routing and Internet connectivity.

Q4. Error Control in Data Link and Transport Layers

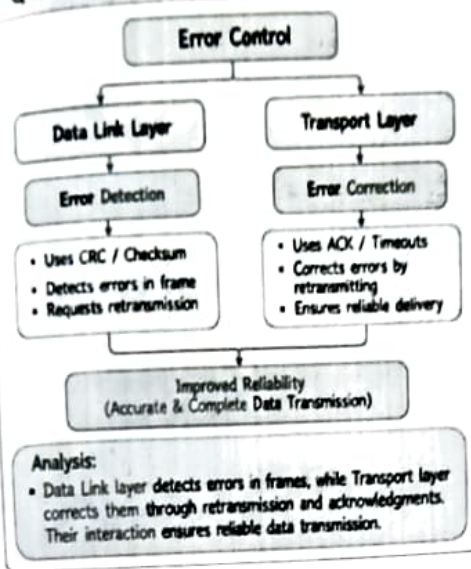
Reliable data transmission is achieved by the coordinated operation of the **Data Link Layer** and the **Transport Layer**, both of which contribute to **Error Control** but at different levels of communication.

The **Data Link Layer** provides **Error Detection** for communication between devices on the same network. It uses techniques such as **CRC (Cyclic Redundancy Check)**, **checksum**, and **parity bits** to detect errors that occur during frame transmission. If an error is detected, the corrupted frame is discarded or retransmitted, preventing damaged data from progressing further through the network.

The **Transport Layer** provides **Error Correction** and **end-to-end reliability** between the source and destination systems. Protocols such as **TCP** use sequence numbers, acknowledgments (ACKs), timers, flow control, and retransmission mechanisms to identify lost or corrupted packets and ensure that all data is delivered correctly and in the proper order.

The interaction between these layers significantly improves communication reliability. The **Data Link Layer** detects and handles errors locally on each network link, reducing the number of corrupted frames forwarded across the network. The **Transport Layer** then verifies complete end-to-end delivery and corrects any remaining transmission errors through retransmission and recovery mechanisms. Together, these layers ensure **accurate, reliable, and efficient data transmission** across both local and global networks.

Q4. Error Control in Data Link and Transport Layers



Layer / Error Control Function	Technique / Mechanism	Role in Error Control	Contribution to Reliability
Data Link Layer	Error Detection	CRC (Cyclic Redundancy Check), Checksum, Parity Bit	Detects transmission errors in frames on a single network link and prevents corrupted frames from being forwarded.
Transport Layer	Error Correction	TCP Sequence Numbers, Acknowledgments (ACKs), Retransmission, Timers	Detects lost or corrupted packets, retransmits them, and ensures complete end-to-end data delivery in the correct order.
Combined Operation	Error Detection + Error Correction	Link-level detection and end-to-end recovery	Improves overall communication reliability by detecting errors locally and correcting them across the entire network path, ensuring accurate and dependable data transmission.

Literature Insights & Evidence (Q4)

- The Data Link Layer improves communication reliability by detecting transmission errors using techniques such as CRC, checksum, and parity checking.
- The Transport Layer, particularly through the TCP protocol, ensures end-to-end reliability using sequence numbers, acknowledgments (ACKs), and retransmissions.

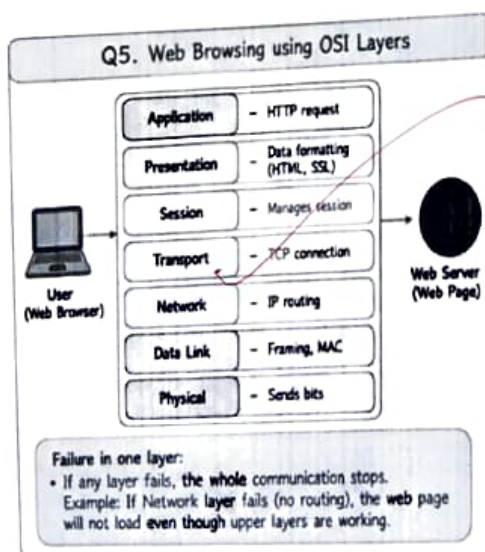
- Research on the OSI model shows that combining link-level error detection with transport-level error correction significantly reduces data loss and improves network performance.
- Standard networking references (e.g., Forouzan and Tanenbaum) emphasize that cooperation between the Data Link and Transport layers is essential for accurate, reliable, and efficient data communication.

Q5. Web Browsing using OSI Layers

The **OSI (Open Systems Interconnection) model** is a seven-layer framework that standardizes how data is transmitted between devices over a network. Each layer performs a specific function, from physical data transmission to user-level applications, ensuring reliable, secure, and efficient communication.

During **web browsing**, data passes through all seven OSI layers. The browser generates an **HTTP/HTTPS** request at the **Application Layer**, the **Presentation Layer** encrypts the data using **SSL/TLS**, the **Session Layer** manages the communication session, the **Transport Layer** (TCP) ensures reliable data delivery, the **Network Layer** routes packets using IP addresses, the **Data Link Layer** transfers frames within the local network, and the **Physical Layer** transmits the data as electrical, optical, or wireless signals.

The layered architecture improves **communication effectiveness** by providing reliability, security, error detection, scalability, and interoperability. Since each layer depends on the services of the lower layer, a failure in any one layer can disrupt the communication process and prevent successful web browsing.



OSI Communication Architecture	Communication Effectiveness Vector	Web Browsing Process & Failure Impact
Application Layer - HTTP/HTTPS, Browser, DNS request generation	Usability - Enables users to access web services correctly	Displays webpages and sends browser requests. Failure: Browser cannot request or display webpages.
Presentation Layer - SSL/TLS encryption, compression, data formatting	Security & Compatibility - Ensures secure and readable data	Encrypts and decrypts HTTPS traffic. Failure: Secure websites cannot establish trusted connections.
Session Layer - Session establishment, maintenance, synchronization	Reliability - Maintains continuous communication	Manages user sessions and logins. Failure: Frequent session disconnections and login interruptions.
Transport Layer - TCP, segmentation, port management, error recovery	Accuracy & Reliability - Ensures error-free and ordered delivery	Delivers webpage data reliably. Failure: Packet loss, incomplete webpages, and connection timeouts.
Network Layer - IP addressing and routing	Scalability - Routes packets across networks	Finds the destination web server. Failure: Packets cannot reach the destination website.
Data Link Layer - MAC addressing, framing, error detection	Integrity - Provides reliable local network communication	Transfers frames between devices on the local network. Failure: Frame corruption and repeated retransmissions.
Physical Layer - Cables, Wi-Fi, optical fiber, electrical signals	Availability - Provides physical connectivity	Transmits bits over the communication medium. Failure: No network connection; communication stops completely.

Literature Insights & Evidence (Q5)

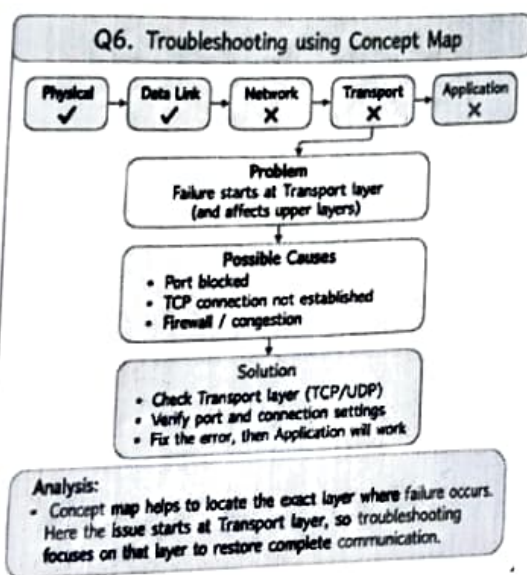
- The OSI model provides a standardized seven-layer framework that improves interoperability, modularity, and efficient communication between different networking systems and devices.
- Research and networking literature highlight that protocols such as TCP/IP, HTTP/HTTPS, DNS, and SSL/TLS work across different OSI layers to ensure reliable, secure, and accurate web browsing.

Q6. Troubleshooting using Concept Map

The concept map shows that the **Physical** and **Data Link** layers are functioning correctly, confirming that the device is connected to the network and frames are transmitted successfully. However, the **Network Layer** has failed due to issues such as incorrect IP addressing, routing errors, or router malfunction. Since the Network Layer is responsible for forwarding packets between networks, its failure prevents data from reaching the destination.

As a result, the **Transport Layer** cannot establish TCP or UDP communication, causing reliable data transfer to fail. Consequently, the **Application Layer** services such as web browsing, email, and file sharing also stop working.

The concept map helps in troubleshooting by showing the dependency between OSI layers. Since the first two layers are operational, engineers can focus directly on the **Network Layer**, reducing troubleshooting time. By checking IP configuration, routing tables, gateway settings, and router connectivity, the fault can be identified and corrected. Once the Network Layer is restored, the Transport and Application layers automatically resume normal operation. This layered approach makes network fault diagnosis faster, systematic, and more efficient.



Concept	Connected To	Purpose / Description
Network Communication	OSI Model	Enables communication between devices.
OSI Model	Physical Layer	Organizes communication into layers for easier management and troubleshooting.
Physical Layer	Data Link Layer	Transmits bits through cables, Wi-Fi, and network hardware.
Data Link Layer	Network Layer	Transfers frames using MAC addresses and detects errors.
Network Layer	Transport Layer	Performs IP addressing and routing. Network issue occurs here.
Transport Layer	Application Layer	Provides reliable end-to-end communication using TCP/UDP.
Application Layer	User Services	Supports applications such as Web, Email, FTP, and DNS.

Network Issue	Troubleshooting	Check IP address, subnet mask, default gateway, routing table, and router configuration.
Troubleshooting	Restored Network	Resolves the issue and restores normal communication.

Literature Insights & Evidence (Q6)

- **The OSI model provides a structured framework for troubleshooting** by dividing network communication into seven layers, enabling network administrators to isolate and resolve faults more efficiently (Zimmermann, 1980).
- **Research indicates that most network issues can be diagnosed faster using a layer-by-layer approach**, as failures in one layer often affect the operation of higher layers, improving troubleshooting accuracy (Forouzan, *Data Communications and Networking*).
- **Industry standards from ISO and Cisco recommend the OSI model as a best practice** for network design, maintenance, and troubleshooting because it simplifies fault isolation and improves interoperability between different networking devices.

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